SmolVLM

Redefining Small and Efficient Multimodal Models

Sub-1 GB inference · 13-benchmark competitive · Fully open-source

Hugging Face

Stanford University

256M

parameters

< 1 GB RAM

on-device inference

300×

larger Idefics-80B outperformed

arxiv.org/abs/2504.05299

Large VLMs Are Powerful but Impractical for Edge Deployment

The Deployment Gap

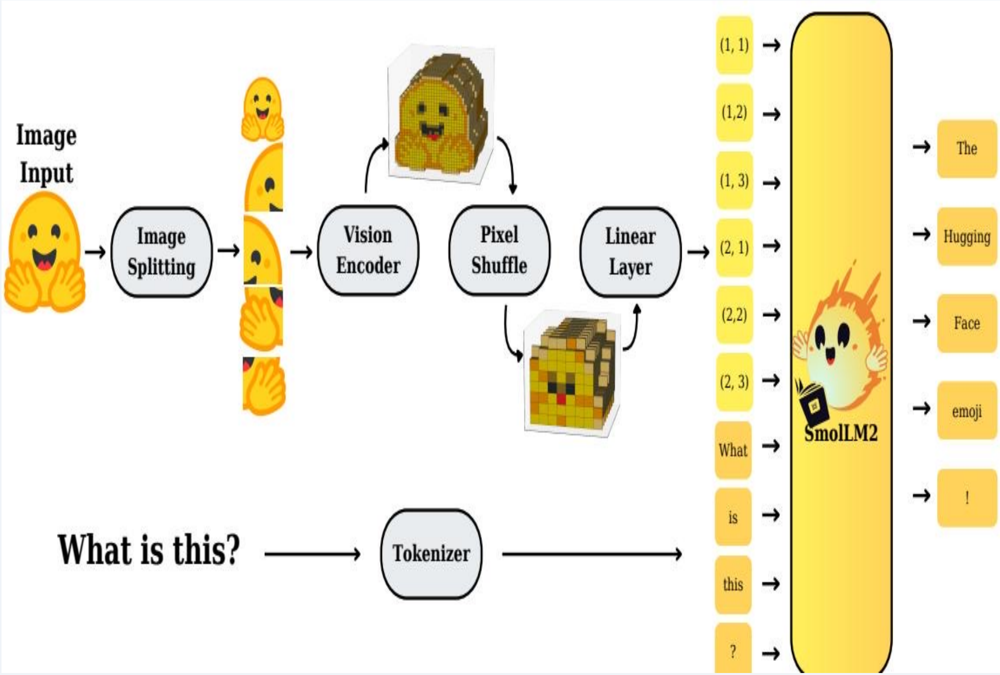
- State-of-the-art VLMs (GPT-4V, LLaVA-34B) require 20–80 GB GPU VRAM
- Mobile phones and edge devices typically have 4–16 GB total RAM
- Cloud inference adds latency, cost, and privacy concerns
- Prior compact VLMs traded accuracy for efficiency without principled design
- No principled framework for scaling-down multimodal architectures

GPU RAM Required at Inference

GPT-4V / Large VLMs	~80 GB
Molmo-A1B-7B (Efficient OS)	27.7 GB
SmolVLM-2.2B	4.9 GB
SmolVLM-256M	0.8 GB

SmolVLM closes this gap through architectural innovation — not just scaling down.

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone



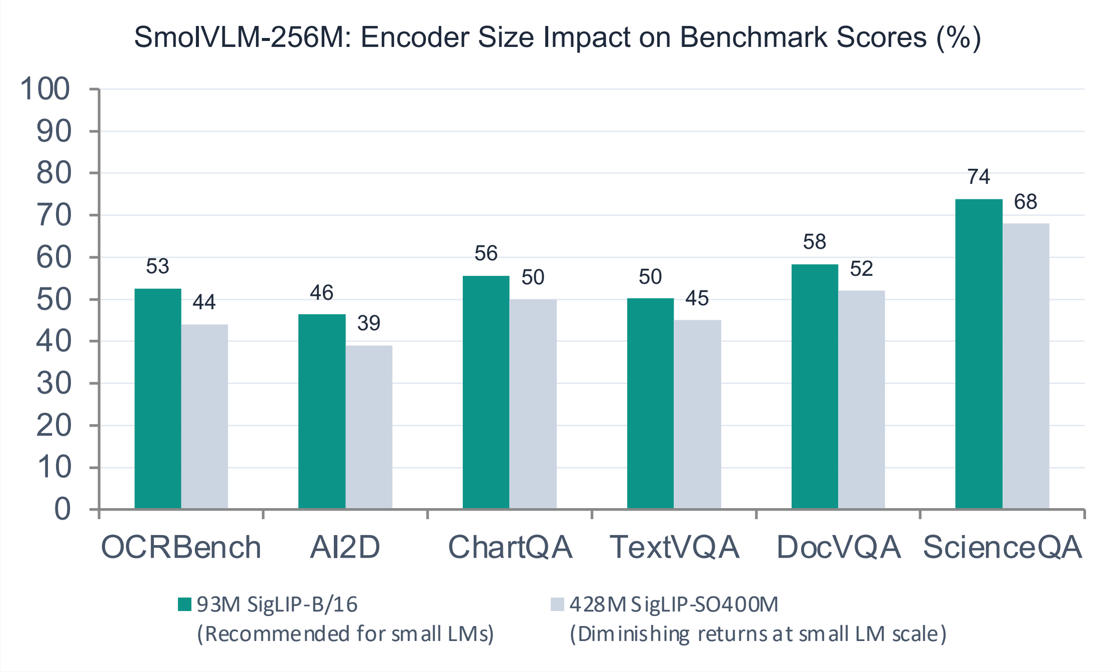
Video frames are sampled and processed through the same pipeline.

Pipeline Steps

1. **Image Splitting**
   Input image divided into sub-images for higher effective resolution
2. **Vision Encoder**
   SigLIP encodes each sub-image patch into visual features
3. **Pixel Shuffle**
   Space-to-depth compression reduces token count by r^2
4. **Linear Projection**
   Projects visual tokens into SmolLM2 embedding space
5. **SmolLM2 LM**
   Fused visual + text tokens processed by language model

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1 — Balanced Encoder-LM Compute: 93M SigLIP-B/16 + 135M LM outperforms 428M SigLIP-SO400M + 135M LM. Larger encoder only justified at ≥1.7B LM scale.



Key Insights

93M

SigLIP-B/16 parameters — optimal for compact VLMs

428M

SigLIP-SO400M — 4.6× larger, underperforms at small LM scale

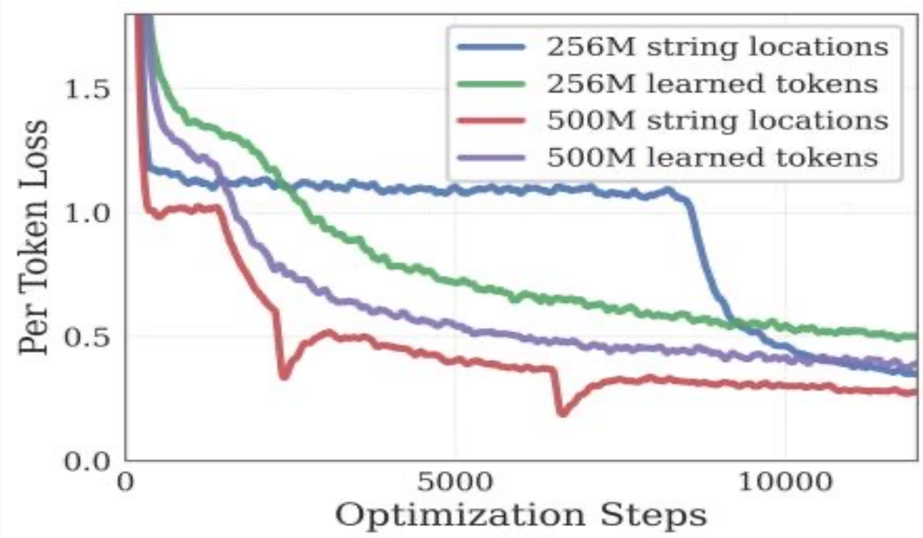
+10%

Parameter overhead of large encoder at 1.7B+ LM scale — only then justified

↑ Perf

Smaller encoder frees capacity for the language model

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models



Pixel Shuffle (Space-to-Depth)

Token count reduced by r^2: at r=4, 256 tokens → 16 tokens, easing attention overhead.

Token Reduction by Shuffle Factor

r=1 1× reduction

256 tokens · No compression (baseline)

r=2 **4× reduction**

64 tokens · Preferred for 1.7B+ models

r=4 **16× reduction**

16 tokens · Optimal for small VLMs (<500M)

Finding 3 — Larger models prefer r=2 (4× reduction) while small VLMs benefit most from r=4 (16× reduction), easing attention overhead and improving long-context modeling.

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2 — Extended Context: Performance improves significantly when extending context from 2K to 16K tokens, enabling higher resolution image understanding and longer video sequences.

Context Extension Strategy

Baseline: 2K context

RoPE base = 10,000

Extended: 8K context

Smaller SmolVLM variants (256M, 500M)

Extended: 16K context

SmolVLM-2.2B · RoPE base = 273,000

Implementation

RoPE Base Adjustment

Increased from 10K → 273K to handle long-range positional dependencies at extended context lengths.

Mixed Data Fine-tuning

Fine-tuned on a blend of long-context and short-context examples to maintain performance across sequence lengths.

Resolution Benefit

Longer context = more image sub-patches = higher effective visual resolution without changing the encoder.

Model-Specific Configs

256M & 500M → 8K context; 2.2B → 16K context, matching compute budget.

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Capability	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS (Baseline)	Baseline Model
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7%	Molmo-A1B-7B
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0%	Molmo-A1B-7B
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0%	Molmo-A1B-7B
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7%	Molmo-A1B-7B
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9%	Molmo-A1B-7B
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6%	Molmo-A1B-7B
Video	Video-MME	33.7%	42.2%	52.1%	45.0%	InternVL2-2B
AVERAGE	**13 Benchmarks**	**44.0%**	**51.0%**	**59.8%**	—	

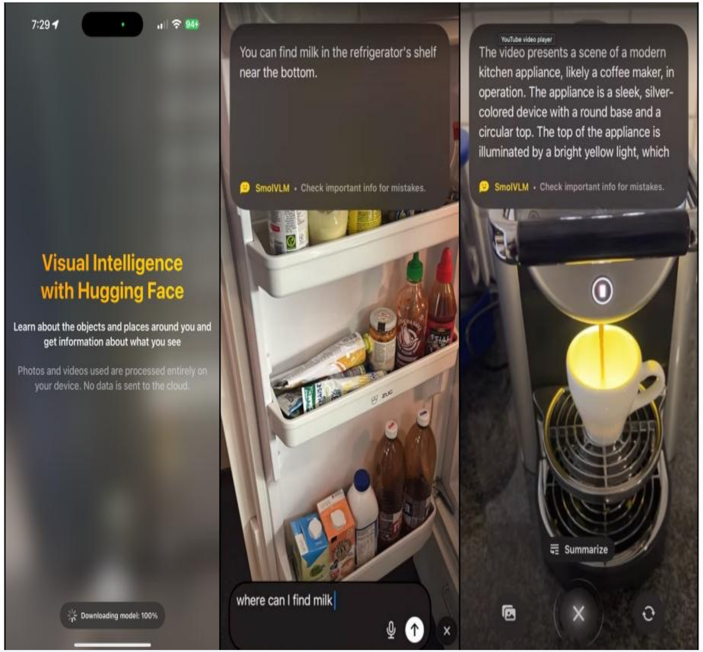
0.8 GB RAM
SmolVLM-256M
batch=1

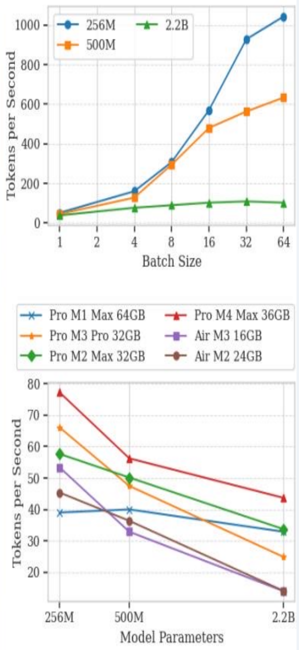
1.2 GB RAM
SmolVLM-500M
batch=1

4.9 GB RAM
SmolVLM-2.2B
batch=1

27.7 GB RAM
Molmo-A1B-7B
(Best OS baseline)

Real-Time Inference on Apple Silicon: Up to 80 Tokens/Second





HuggingSnap: On-Device Results

Processes photos & videos locally on iOS — no cloud dependency.

~80 tok/s
MacBook Pro M1 Max 64GB
SmolVLM-256M, batch=1

~55 tok/s
MacBook Air M3 16GB
SmolVLM-256M, batch=1

~1000 tok/s
High-end (batch=64)
SmolVLM-256M

Throughput scales ~linearly with batch size for the 256M model.

Key Takeaways

Design Principles for Efficient Multimodal Models

01 Balance Encoder-LM Compute

Use 93M SigLIP-B/16 for models <500M. Larger encoders only justify their overhead at ≥1.7B LM scale.

02 Extend Context Length Aggressively

2K → 16K context (RoPE base 10K → 273K) + mixed long/short fine-tuning yields significant accuracy gains.

03 Maximize Token Compression for Small Models

r=4 pixel shuffle delivers 16× token reduction, cutting attention overhead and enabling efficient long-context modeling.

04 Sub-1GB On-Device Inference Is Achievable

SmolVLM-256M runs at 0.8 GB RAM, ~80 tok/s on Apple Silicon, outperforming the 300× larger Idefics-80B.

05 Open Science Accelerates Edge AI

Full open-source release (weights, data, code, HuggingSnap app) enables reproducible on-device AI research.